

Transformations of reciprocal trigonometric functions



1 Consider the function $f(x) = \csc x$, which is defined as the reciprocal function of $\sin x$.

- a Describe the values of x where $\sin x = 0$.
- b What is true about the function $f(x) = \csc x$ at these x -values?

2 Consider the function $f(x) = \sec x$, which is defined as the reciprocal function of $\cos x$.

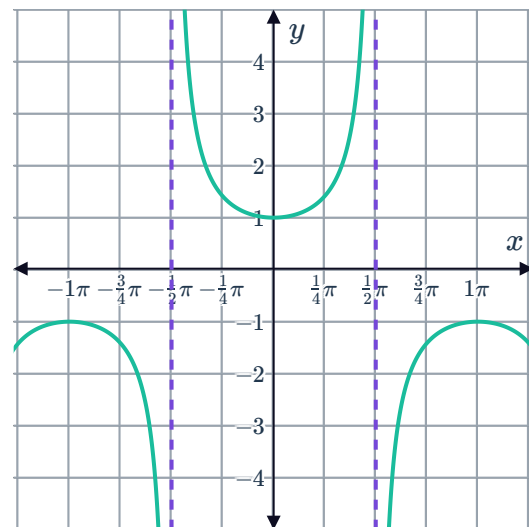
- a Describe the values of x where $\cos x = 0$.
- b What is true about the function $f(x) = \sec x$ at these x -values?

3 Consider the function $f(x) = \cot x$, which is defined as the reciprocal function of $\tan x$.

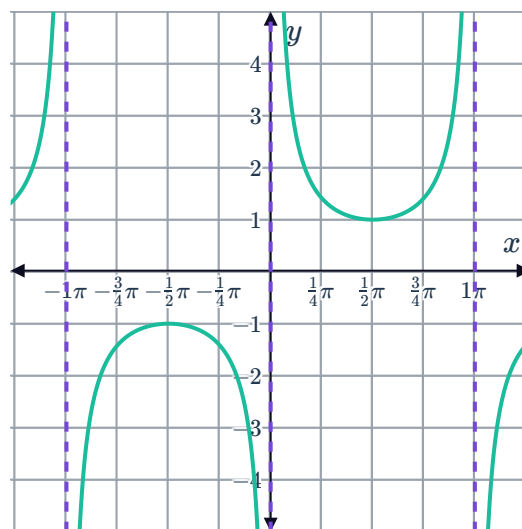
- a Describe the values of x where $\tan x = 0$.
- b What is true about the function $f(x) = \cot x$ at these x -values?

4 Consider the graph of $y = \sec x$ and its first local minimum for $x \geq 0$ at $(0, 1)$.
State the coordinates of the first local minimum of each of the given functions for $x \geq 0$:

- a $y = \sec 3x$
- b $y = \sec\left(x - \frac{\pi}{4}\right)$
- c $y = 5 \sec x + 2$



- 5 Consider the graph of $y = \csc x$ and its first local minimum for $x \geq 0$ at $\left(\frac{\pi}{2}, 1\right)$.
State the coordinates of the first local minimum of each of the given functions for $x \geq 0$:

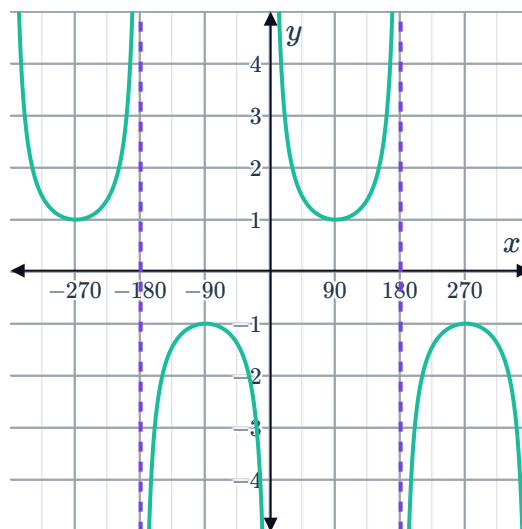


a $y = 5 \csc x$ b $y = -5 \csc x$

c $y = \csc x + 2$

- 6 Consider the graph of the function $y = \csc x$:

- a Sketch the graph of $y = \csc(x + 90^\circ)$.
b Now, sketch the graph of $y = -2 \csc(x + 90^\circ)$.



- 7 Consider the function $y = \csc x$.

- a Sketch the graph of the function $y = \csc\left(x + \frac{3\pi}{4}\right)$ for $-\pi \leq x \leq \pi$.
b Now, sketch the graph of the function $y = \frac{3}{2} \csc\left(x + \frac{3\pi}{4}\right)$ for $-\pi \leq x \leq \pi$.

- 8 Sketch the graph of the following functions for $-2\pi \leq x \leq 2\pi$:

a $y = 2 \sec\left(2x - \frac{\pi}{2}\right) - 2$

b $y = 3 \csc\left(\frac{x}{2} - \pi\right) - 1$

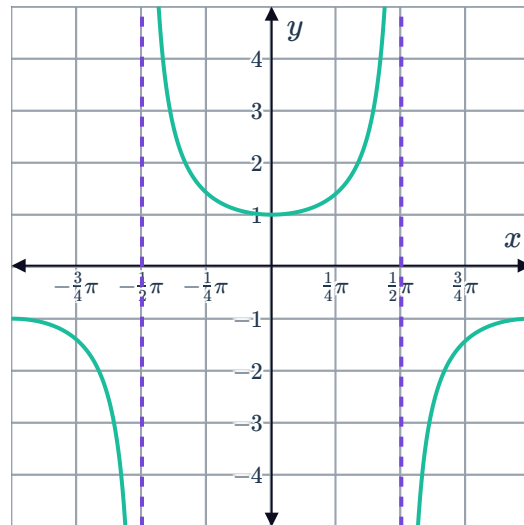
c $y = 2 \cot\left(\frac{x}{2} - \frac{\pi}{2}\right) - 1$

- 9 Consider the function $y = \cot \left(3 \left(x - \frac{\pi}{6} \right) \right)$ 

Describe the series of transformations to get the graph of $y = \cot \left(3 \left(x - \frac{\pi}{6} \right) \right)$ from $y = \cot x$.

- 10 Consider the graph of $y = \sec x$.

- Describe the transform from the graph of $y = \sec x$ to create the graph of $y = \sec \left(x - \frac{\pi}{2} \right) - 2$.
- Now, sketch the graph of the function $y = \sec \left(x - \frac{\pi}{2} \right) - 2$ on the same coordinate plane.
- Find the period of $y = \sec \left(x - \frac{\pi}{2} \right) - 2$.



Domain and range of reciprocal trigonometric functions

- 11 Consider the function $f(x) = \sec x + 3$.

- Describe the transformations of $g(x) = \sec x$ that will produce $f(x)$.
- Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - Vertical asymptotes
 - Relative minima and maxima
 - Range
 - Domain
- Now, find the domain of $f(x)$ in interval notation.
- State the range of $f(x)$ in interval notation.

12 Consider the function $f(x) = \csc x + 2$.



- a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Range
 - ii Domain
 - iii Relative minima and maxima
 - iv Vertical asymptotes
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

13 Consider the function $f(x) = \csc\left(x + \frac{3\pi}{4}\right)$.

- a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Range
 - ii Domain
 - iii Vertical asymptotes
 - iv Period
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

14 Consider the function $f(x) = \sec\left(x - \frac{3\pi}{4}\right)$.

- a Describe the transformation of $g(x) = \sec x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Domain
 - ii Vertical asymptotes
 - iii Range
 - iv Period
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

15 Consider the function $f(x) = \cot\left(x - \frac{3\pi}{4}\right)$



- a Describe the transformation of $g(x) = \cot x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Domain
 - ii Range
 - iii Vertical asymptotes
 - iv Period
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

16 Consider the function $f(x) = \csc 4x$.

- a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Vertical asymptotes
 - ii Range
 - iii Domain
 - iv Period
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

17 Consider the function $f(x) = \cot\left(\frac{x}{3}\right)$.

- a Describe the transformation of $g(x) = \cot x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Domain
 - ii Range
 - iii Vertical asymptotes
 - iv Period
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

18 Consider the function $f(x) = \sec 3x$.



- a Describe the transformation of $g(x) = \sec x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Period
 - ii Domain
 - iii Range
 - iv Vertical asymptotes
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

19 Consider the function $f(x) = 3 \cot x$.

- a Describe the transformation of $g(x) = \cot x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Domain
 - ii Range
 - iii Vertical asymptotes
 - iv Period
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

20 Consider the function $f(x) = 4 \sec x$.

- a Describe the transformation of $g(x) = \sec x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Vertical asymptotes
 - ii Period
 - iii Range
 - iv Domain
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

21 Consider the function $f(x) = 5 \csc x$.



- a Describe the transformation of $g(x) = \csc x$ that will produce $f(x)$.
- b Determine whether each of the following properties will change when transforming $g(x)$ to get $f(x)$:
 - i Period
 - ii Vertical asymptotes
 - iii Range
 - iv Domain
- c Now, find the domain of $f(x)$ in interval notation.
- d State the range of $f(x)$ in interval notation.

22 Consider the function $f(x) = \csc\left(x + \frac{1}{4}\pi\right) - 6$.

- a State the period of the function.
- b For what values of x are there vertical asymptotes on the graph of $y = f(x)$?
- c Find the coordinates of the relative maxima and relative minima in the domain $(0, 2\pi)$.
- d Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$.
- e Find the domain of $f(x)$ in interval notation.
- f Find the range of $f(x)$ in interval notation.

23 Consider the function $f(x) = \cot\left(x + \frac{1}{4}\pi\right) - 6$.

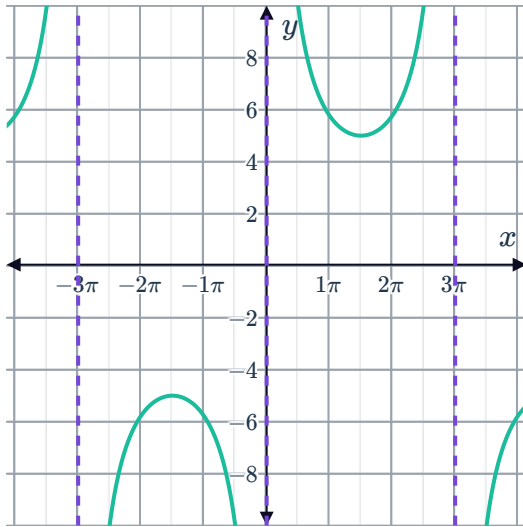
- a State the period of the function.
- b For what values of x are there vertical asymptotes on the graph of $y = f(x)$?
- c Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$.
- d Find the domain of $f(x)$ in interval notation.
- e Find the range of $f(x)$ in interval notation.

24 For each of the following functions:

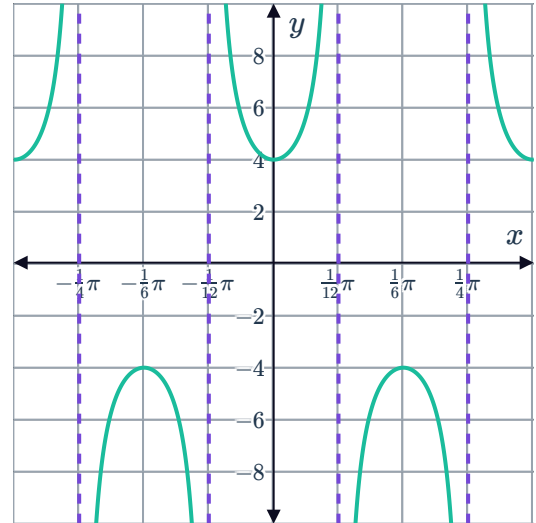


- i Determine the period of the function.
- ii Determine the horizontal translation (phase shift) of the function.
- iii State the range of the function.

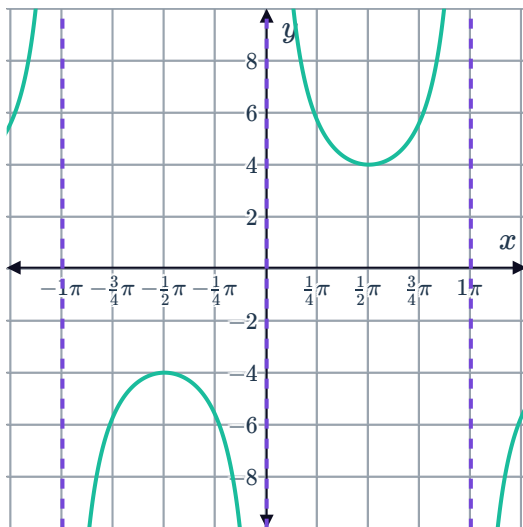
a $y = 5 \csc\left(\frac{1}{3}x\right)$



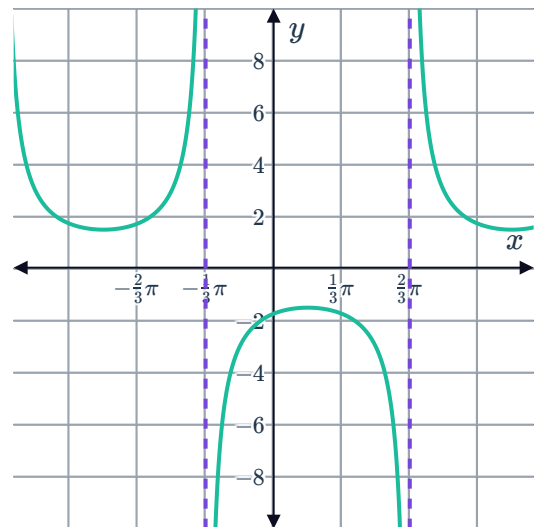
b $y = 4 \sec(6x)$



c $y = -4 \sec\left(x + \frac{\pi}{2}\right)$



d $y = -\frac{3}{2} \csc\left(x - \frac{\pi}{6}\right)$



25

Consider the function $f(x) = \sec\left(x + \frac{1}{4}\pi\right)$



- State the period of the function.
- For what values of x are there vertical asymptotes on the graph of $y = f(x)$?
- Find the coordinates of the relative maxima and relative minima in the domain $(0, 2\pi)$.
- Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$.
- Find the domain of $f(x)$ in interval notation.
- Find the range of $f(x)$ in interval notation.

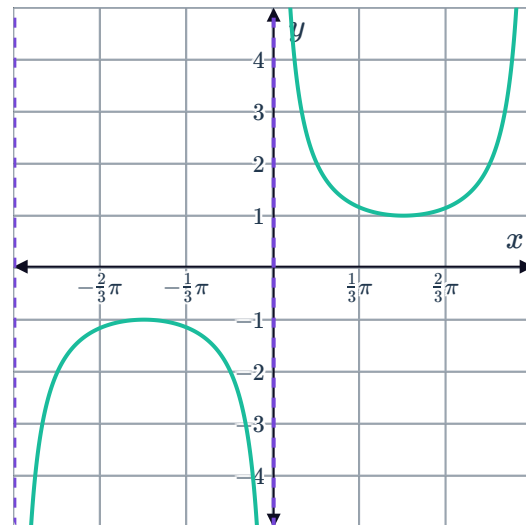
Equations of reciprocal trigonometric functions

26 For each of the following functions, determine the equation of the new function after undergoing the series of transformations:

- The curve $y = \csc x$ is reflected over the x -axis and translated 3 units down.
- The curve $y = \sec x$ is translated $\frac{\pi}{2}$ units to the left, and then reflected over the x -axis.

27 The graph of $y = \csc x$ shown is stretched vertically by a scale factor of 2 and translated horizontally $\frac{\pi}{3}$ units to the left.

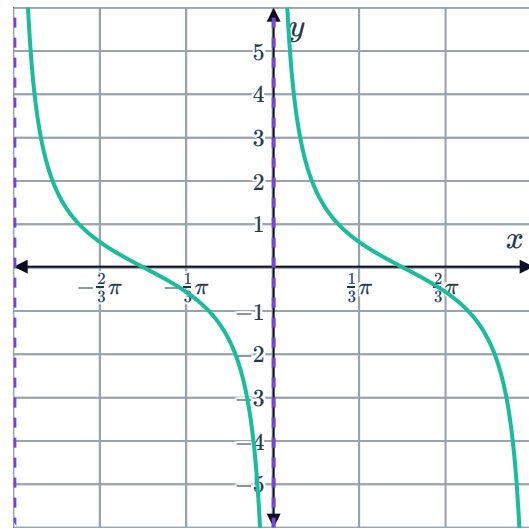
- Find the equation of the new graph.
- Sketch the graph of the transformed function.



- 28** The graph of $y = \cot x$ shown is stretched vertically by a scale factor of 5 and translated horizontally $\frac{\pi}{6}$ units to the right.

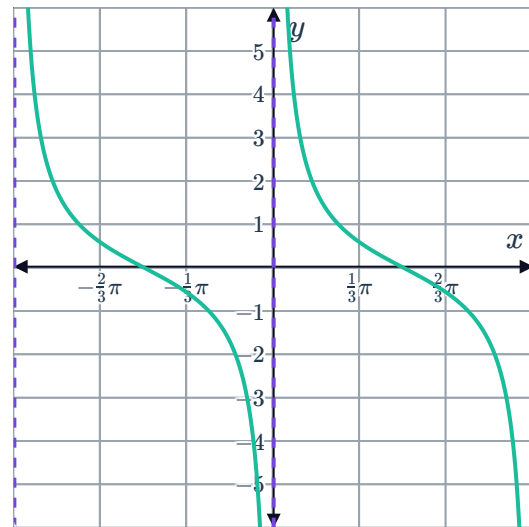


- Find the equation of the new graph.
- Sketch the graph of the transformed function.



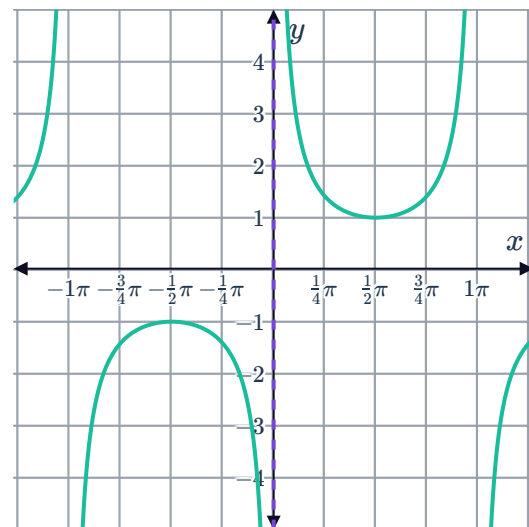
- 29** The graph of $y = \cot x$ shown is compressed horizontally by a scale factor of $\frac{1}{3}$ and then reflected over the x -axis.

- Find the equation of the new graph.
- Sketch the graph of the transformed function.



- 30** Consider the following graph:

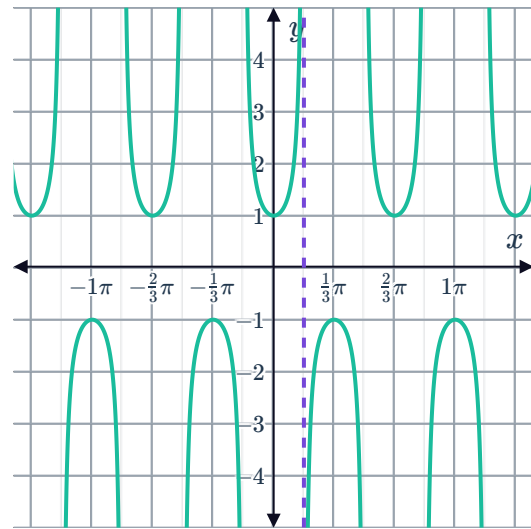
- Find the equation of the asymptote shown.
- Which key feature occurs at the point where $x = \frac{\pi}{2}$?
- Find the period of this function.
- Find the equation of this function.



31 Consider the following graph:



- Find the equation of the asymptote shown.
- Which key feature occurs at the point where $x = \frac{\pi}{3}$?
- Find the period of this function.
- Find the equation of this function.



32 Consider the following graph:

- Find the equation of the asymptote shown.
- Find the period of this function.
- Find the median value of this function.
- Find the equation of this function.

